

14.4 Truncated Singular Value Decomposition



The core skills you'll master in this section:

- ▶ Truncated SVD: keeps the principal components and discards noise and redundant information.
- ▶ Decomposes the data matrix into a sum of multiple rank-one matrices.
- ▶ Loading = eigenvector, score = projection coordinate.
- ▶ Low-rank approximation reconstructs the original data: the first p principal components efficiently reconstruct the data while compressing storage.
- ▶ X is not orthogonal in the original space, but V 's projection result Z achieves orthogonalization.

Earlier in this chapter we introduced the four types of SVD; this section expands on truncated SVD. Truncated SVD is especially widely applied — for example, in principal component analysis, dimensionality reduction, and more.

The Second Perspective of Matrix Multiplication

Let X be the original data matrix, with shape $n \times D$. Its economy SVD is:

$$X_{n \times D} = U_{n \times D} S_{D \times D} V_{D \times D}^T$$

where U and X have the same shape; U is a semi-orthogonal matrix (its column vectors are unit vectors and pairwise orthogonal); V is an orthogonal matrix (its column vectors are unit vectors and pairwise orthogonal).

As shown in Figure 1, from the second perspective of matrix multiplication, the original data matrix X 's economy SVD can be expanded to give

$$X_{n \times D} = \underbrace{\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_D \end{bmatrix}}_{U_{n \times D}} \underbrace{\begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_D \end{bmatrix}}_{S_{D \times D}} \underbrace{\begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_D^T \end{bmatrix}}_{V_{D \times D}^T} = s_1 \mathbf{u}_1 \mathbf{v}_1^T + s_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + s_D \mathbf{u}_D \mathbf{v}_D^T = \sum_{j=1}^D s_j \mathbf{u}_j \mathbf{v}_j^T$$

This way, we can write the data matrix X as the sum of D identically shaped matrices, specifically as shown in Figure 2.

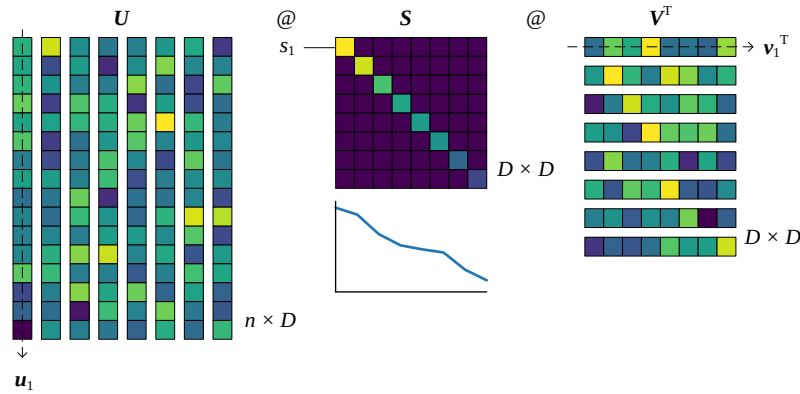


Figure 1. Expanding the Economy SVD via the Second Perspective of Matrix Multiplication

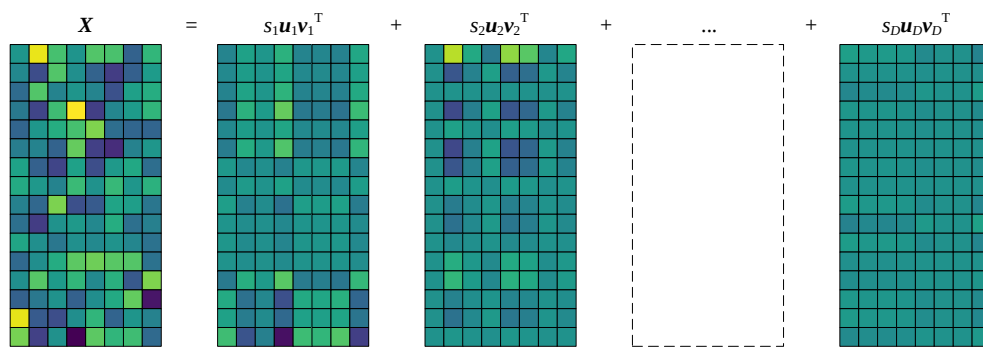


Figure 2. The Original Data is Equivalent to the Sum of D Identically Shaped Matrices

Singular Values Reflect Importance

As shown in Figure 3 and Figure 4, since u_j and v_j are all unit vectors — that is, their vector L^2 norms are all 1.

These unit vectors differ from each other only in direction, not in magnitude. Therefore, the magnitude of the singular value s_j reflects the importance of that direction.

The diagonal square matrix S 's main-diagonal elements act, in one view, by scaling U 's column vectors; in another view, they can be seen as scaling V^T 's row vectors (that is, V 's column vectors).

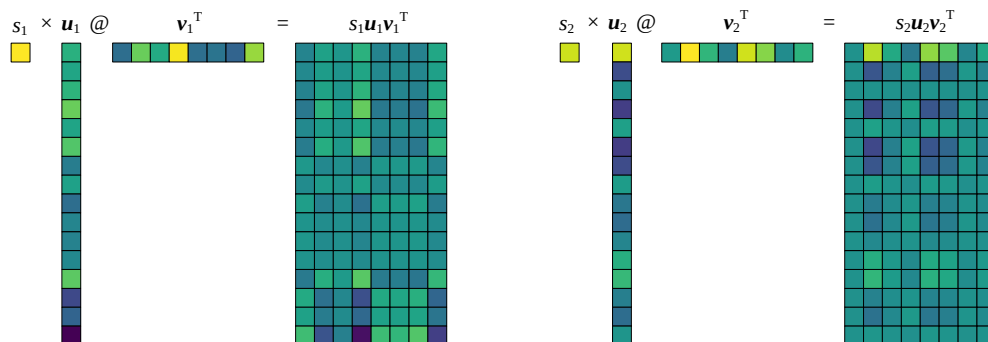


Figure 3. The First Two Principal Components Reconstruct Part of the Original Data

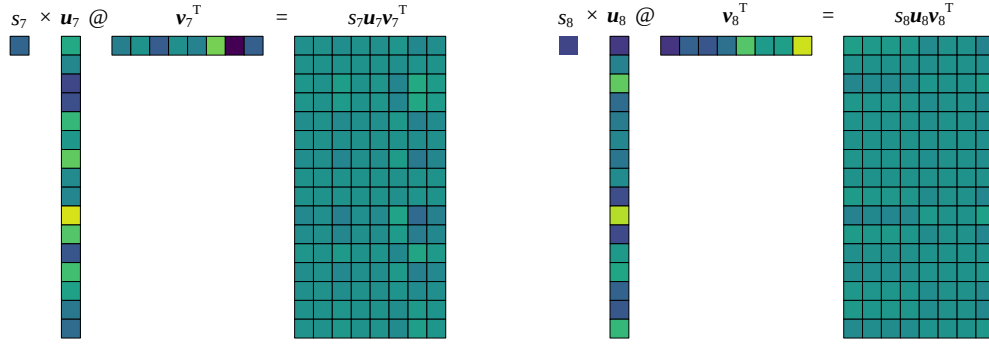


Figure 4. The Last Two Principal Components Reconstruct Part of the Original Data

From the perspective of principal component analysis, if the singular values $s_1, s_2, \dots, s_D$ are arranged from large to small, $\mathbf{v}_1$ is the first principal component's loading, $\mathbf{u}_1$ is the first principal component's factor score, and so on.

In other words, loading = eigenvector, score = projection coordinate.

⚠ Note that some references define loading as "eigenvector $\times$ singular value".

Approximate Reconstruction

As shown in Figure 5, using the first p principal components to reconstruct the original data

$$\mathbf{X}_{n \times D} \approx \hat{\mathbf{X}}_{n \times D} = \mathbf{U}_{n \times p} \mathbf{S}_{p \times p} (\mathbf{V}_{D \times p})^T = \sum_{j=1}^p s_j \mathbf{u}_j \mathbf{v}_j^T$$

Assume the first p singular values s_j are all greater than 0; then matrix $\hat{\mathbf{X}}_{n \times D}$ has rank p .

In other words, provided that s_j are all greater than 0, each additional layer added in the equation above, $s_j \mathbf{u}_j \mathbf{v}_j^T$, $\hat{\mathbf{X}}_{n \times D}$'s rank increases by 1.

⚠ Note that when using different Python functions to compute the SVD, we often find the resulting unit vectors differ in sign; this is because $s_j \mathbf{u}_j \mathbf{v}_j^T$ is equivalent to $s_j (-\mathbf{u}_j)(-\mathbf{v}_j)^T$, meaning $\mathbf{u}_j$ and $\mathbf{v}_j$ can flip sign simultaneously without affecting the result.

For example, $\hat{\mathbf{X}}_{n \times D} = s_1 \mathbf{u}_1 \mathbf{v}_1^T$ has rank 1; $\hat{\mathbf{X}}_{n \times D} = s_1 \mathbf{u}_1 \mathbf{v}_1^T + s_2 \mathbf{u}_2 \mathbf{v}_2^T$ has rank 2.

Figure 5: $\mathbf{X}$ and $\hat{\mathbf{X}}$'s heatmap error matrix is:

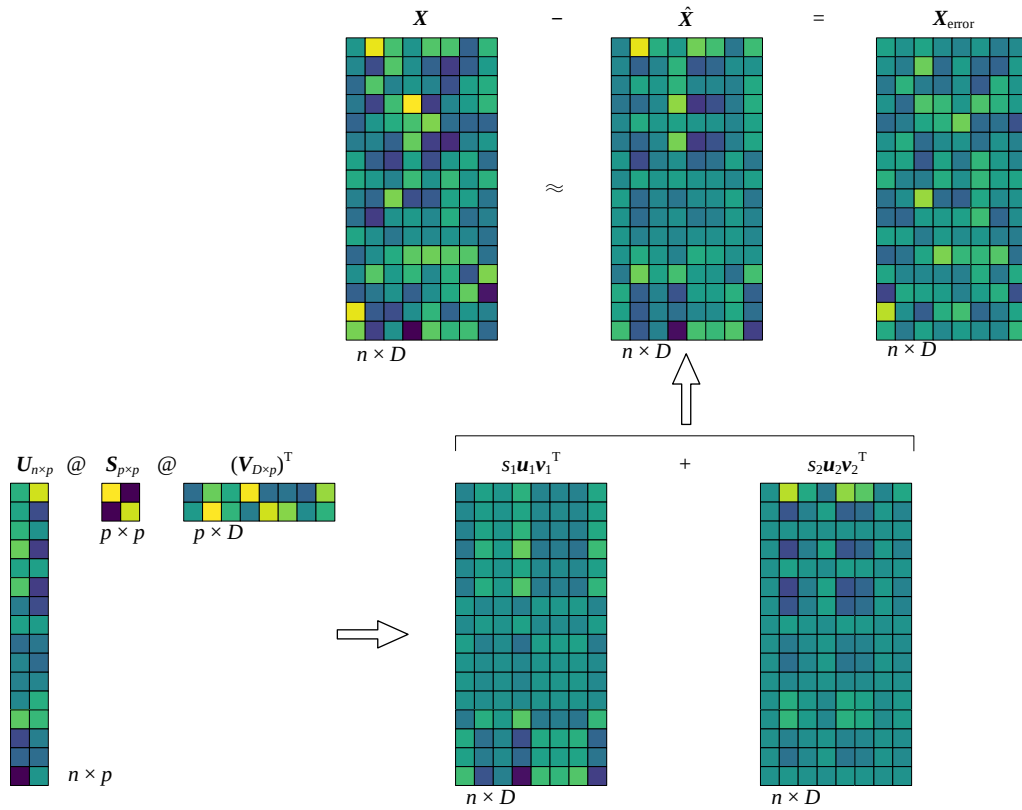
$$\mathbf{X}_{\text{error}} = \mathbf{X}_{n \times D} - \hat{\mathbf{X}}_{n \times D}$$

From the equation above, we know that the error is likewise a sum of several layers, that is,

$$\mathbf{X}_{\text{error}} = \sum_{j=1}^D s_j \mathbf{u}_j \mathbf{v}_j^T - \sum_{j=1}^p s_j \mathbf{u}_j \mathbf{v}_j^T = s_{p+1} \mathbf{u}_{p+1} \mathbf{v}_{p+1}^T + \dots + s_D \mathbf{u}_D \mathbf{v}_D^T = \sum_{j=p+1}^D s_j \mathbf{u}_j \mathbf{v}_j^T$$



Please review the error computation in the principal component analysis of Chapter 12, Section 3 of this book.

Figure 5. Reconstructing the Original Data Using the First p Principal Components

Orthogonal Projection

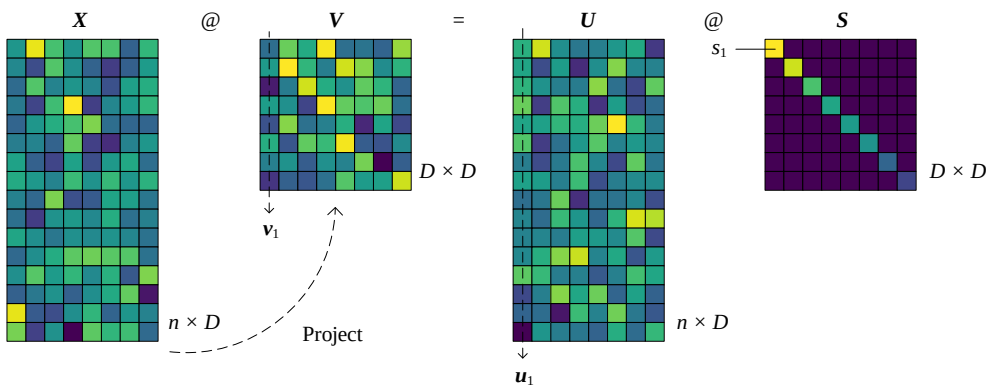
Next, let's understand the truncated SVD from the perspective of projection.

➔ Please review the content on Gram matrix spectral decomposition in Chapter 11, Section 4 of this book.

Write the equation above as

$$X_{n \times D} V_{D \times D} = U_{n \times D} S_{D \times D}$$

As shown in Figure 6, the equation above is equivalent to projecting X onto V 's space.

Figure 6. Projecting the Original Data Onto V

Also using the second perspective of matrix multiplication, write the equation above as

$$\mathbf{X}_{n \times D} \underbrace{[\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_D]}_{\mathbf{V}_{D \times D}} = \underbrace{[\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_D]}_{\mathbf{U}_{n \times D}} \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_D \end{bmatrix}$$

Expanding further gives

$$[\mathbf{X}\mathbf{v}_1 \ \mathbf{X}\mathbf{v}_2 \ \cdots \ \mathbf{X}\mathbf{v}_D] = [s_1\mathbf{u}_1 \ s_2\mathbf{u}_2 \ \cdots \ s_D\mathbf{u}_D]$$

Geometrically, $\mathbf{X}$'s scalar projection onto $\mathbf{v}_j$ is $s_j\mathbf{u}_j$.

$$\mathbf{X}\mathbf{v}_j = s_j\mathbf{u}_j$$

Figure 7 shows the original data's projection onto $\mathbf{v}_1$ with result $s_1\mathbf{u}_1$. Since $\|\mathbf{u}_j\| = 1$, $\|\mathbf{X}\mathbf{v}_j\| = s_j$.

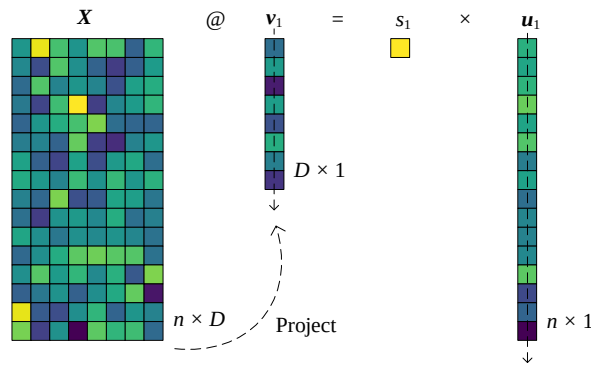


Figure 7. Scalar Projection of the Original Data Onto $\mathbf{v}_1$

As shown in Figure 8, the data matrix $\mathbf{X}$ projected onto the orthonormal basis $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_D]$, which spans a D -dimensional space, can be written as

$$\mathbf{Z} = \mathbf{X}\mathbf{V}$$

$\mathbf{Z}$ represents $\mathbf{X}$'s coordinates in the new orthonormal basis $[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_D]$.

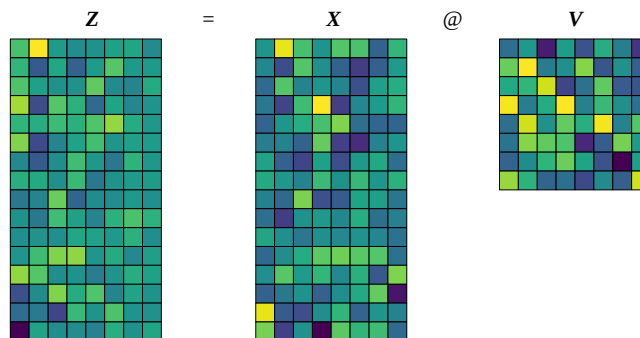


Figure 8. Projecting $\mathbf{X}$ Onto the Orthonormal Basis $\mathbf{V}$

Since $\mathbf{X} = \mathbf{USV}^T$, substituting into the equation above gives:

$$\mathbf{Z} = \mathbf{USV}^T \mathbf{V} = \mathbf{US} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_D] \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_D \end{bmatrix} = [s_1 \mathbf{u}_1 \quad s_2 \mathbf{u}_2 \quad \cdots \quad s_D \mathbf{u}_D]$$

That is,

$$\underbrace{[\mathbf{z}_1 \quad \mathbf{z}_2 \quad \cdots \quad \mathbf{z}_D]}_{\mathbf{Z}} = \underbrace{[s_1 \mathbf{u}_1 \quad s_2 \mathbf{u}_2 \quad \cdots \quad s_D \mathbf{u}_D]}_{\mathbf{US}}$$

As shown in Figure 9, the equation above gives us a second way to compute $\mathbf{Z}$. In other words, $\mathbf{u}_j$ is actually the "normalized" projection coordinate, and s_j is $\mathbf{z}_j$'s vector magnitude — that is, $\|\mathbf{x}_j\| = \|\mathbf{z}_j\| = \|s_j \mathbf{u}_j\| = s_j \|\mathbf{u}_j\| = s_j$.

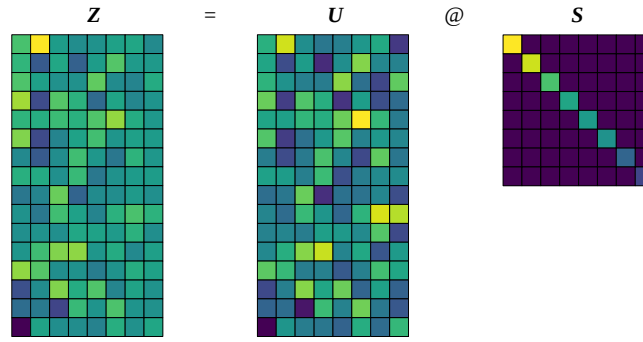


Figure 9. The Second Way to Compute $\mathbf{Z}$

Orthogonalization

Computing the Gram matrix of $\mathbf{Z}$:

$$\mathbf{Z}^T \mathbf{Z} = \begin{bmatrix} \mathbf{z}_1^T \\ \mathbf{z}_2^T \\ \vdots \\ \mathbf{z}_D^T \end{bmatrix} [\mathbf{z}_1 \quad \mathbf{z}_2 \quad \cdots \quad \mathbf{z}_D] = \begin{bmatrix} \mathbf{z}_1^T \mathbf{z}_1 & \mathbf{z}_1^T \mathbf{z}_2 & \cdots & \mathbf{z}_1^T \mathbf{z}_D \\ \mathbf{z}_2^T \mathbf{z}_1 & \mathbf{z}_2^T \mathbf{z}_2 & \cdots & \mathbf{z}_2^T \mathbf{z}_D \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{z}_D^T \mathbf{z}_1 & \mathbf{z}_D^T \mathbf{z}_2 & \cdots & \mathbf{z}_D^T \mathbf{z}_D \end{bmatrix}$$

Please write the equation above in the form of a vector inner product.

Substituting the first equation above into the second equation above gives:

$$\mathbf{Z}^T \mathbf{Z} = (\mathbf{US})^T \mathbf{US} = \mathbf{S}^T \mathbf{U}^T \mathbf{U} \mathbf{S} = \begin{bmatrix} s_1^2 & & & \\ & s_2^2 & & \\ & & \ddots & \\ & & & s_D^2 \end{bmatrix}$$

As shown in Figure 10, we find that $\mathbf{Z}$'s Gram matrix is a diagonal matrix, meaning $\mathbf{Z}$'s column vectors are pairwise orthogonal, that is:

$$\mathbf{z}_i^T \mathbf{z}_j = \mathbf{z}_j^T \mathbf{z}_i = \mathbf{z}_i \cdot \mathbf{z}_j = \mathbf{z}_j \cdot \mathbf{z}_i = \langle \mathbf{z}_i, \mathbf{z}_j \rangle = \langle \mathbf{z}_j, \mathbf{z}_i \rangle = 0, \quad i \neq j$$

Looking back at Figure 8, $X \rightarrow Z$ accomplishes exactly the orthogonalization.

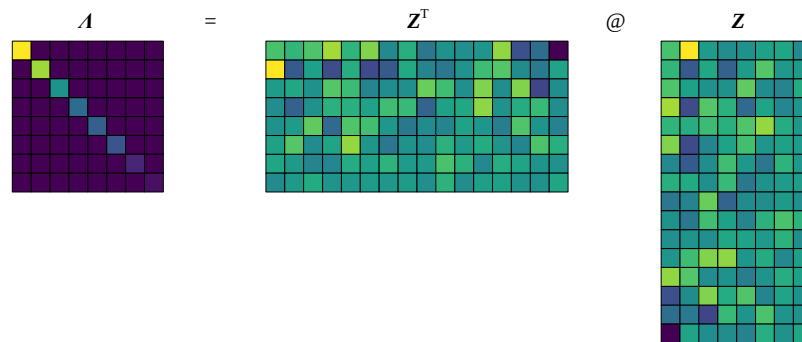


Figure 10. Z 's Gram Matrix

Analyzing an Iris Photo

Next, let's use truncated SVD to analyze an iris photo. Figure 11 shows a photo of an iris taken by the author; after converting it to grayscale, each pixel is a number in the range $[0, 1]$. So the entire image can be viewed as a data matrix.



This part is adapted from "Machine Learning" in the Iris Series.

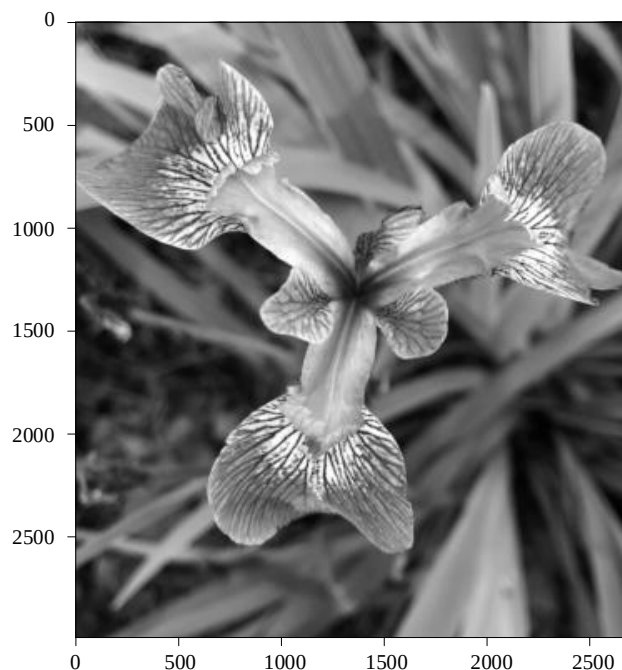


Figure 11. Iris Photo, Converted to Grayscale, from Machine Learning

Figure 12 shows how the singular values obtained from the SVD change with the principal components. The first 10 singular values contribute the most and dominate the data structure.

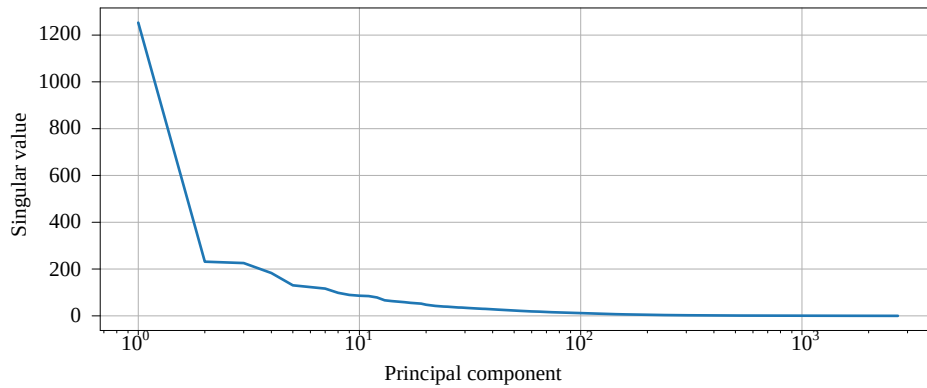


Figure 12. Singular Values vs. Principal Components, from Machine Learning

Figure 13 shows the iris photo reconstructed using the first 4 principal components. This image is equivalent to the sum of 4 rank-one matrices, specifically as shown in Figure 14.

Figure 13's left panel — all we can make out is a "grid" pattern.

Figure 15 shows the iris image reconstructed using the first 64 principal components — the image is already quite clear, and we can already make out the shape of the iris. Compared to the original image, Figure 15's data is compressed dramatically, retaining only about 2.5% (64/2714).

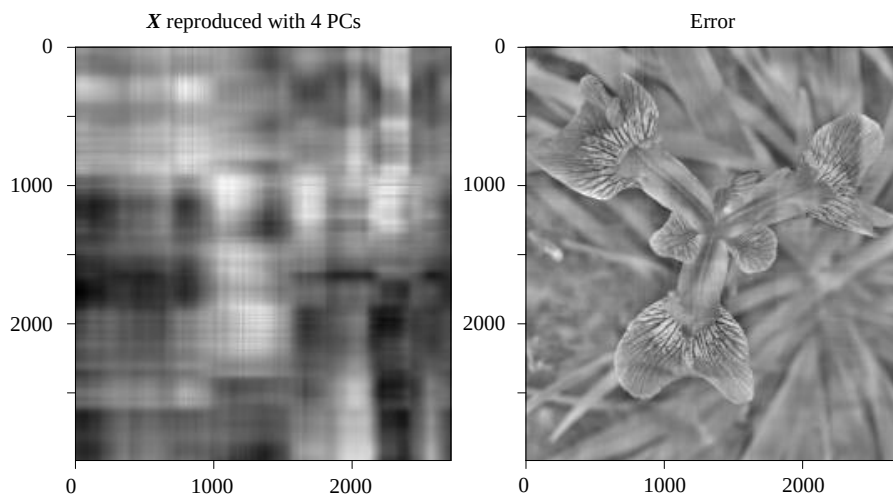


Figure 13. Reconstructing the Iris Photo Using the 1st, 2nd, 3rd, and 4th Principal Components

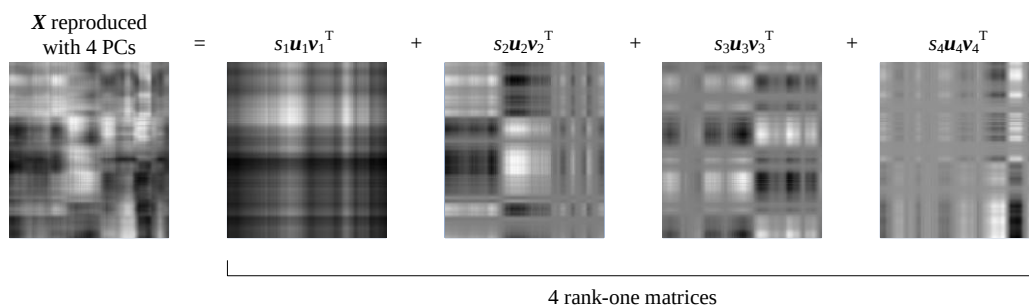


Figure 14. Sum of the First 4 Rank-One Matrices

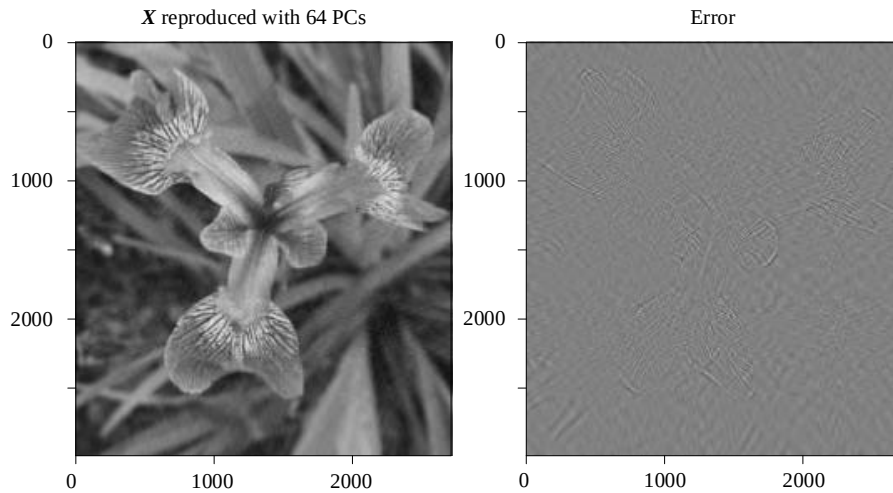


Figure 15. Reconstructing the Iris Photo Using the First 64 Principal Components



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please perform the economy singular value decomposition of the following matrix, then keep the first principal component to approximately reconstruct the original matrix, and compute the error.

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$